

The dynamic sector

Kinetic energy as a ξ -fractal fraction of the static ($\tilde{T} \cdot m = 1$): problem statement, narrowing, and the structural lead

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Abstract

Through the duality $\tilde{T} \cdot m = 1$, FFGFT separates two energy types: *static* (rest mass = topologically frozen winding, the crystallographic Z_3 core) and *kinetic* (free; for massless states the only form, $E = hf$). The static sector is derived and computed (Z_3 mass operator + mass circle; Doc. 285, BD12-XCORE). The dynamic sector is *not*: the derived rule that computes the kinetic part, via the duality, as a ξ -fractal fraction of the static one is missing. This document is a *problem-statement document* – it records what is fixed (the duality places the window in the kinetic sector), the single existing data point (neutrino, $m_\nu \sim \xi^2/2 m_e$, as an ansatz), the narrowing of the ξ -power to $p = 2$, and the structural lead $m_{\text{window}} = \xi \cdot \omega$. What remains to be derived is marked explicitly as open. Prefixed is the fixed, reversible Hilbert-space foundation (Doc. 282/270/230) the sector sits on; its actual aim is to find, out of the Hilbert space, the complementary description that maps via $D_f = 3 - \xi$ and ξ onto SI units.

1 Where the question comes from

In Doc. 255, $\tilde{T} \cdot m = 1$ enforces two energy types. The rest mass is *static*: a topologically stable winding number on T^4 , fixed in the compact picture by the Z_3 mass operator ($r = \sqrt{2}$, $\theta = 2/9$). The kinetic energy is *free*; for a massless particle it is the only form ($E = hf$). "Mass is frozen kinetic energy."

Doc. 285 shows that the empty recursion windows ($3'$, $3''$, ...) carry no *new* masses but structure; and that a *dynamics* filling a window supplies the *kinetic* type – a filled window is photon-/neutrino-like, not a massive triple. There the gap was named: the *derived* rule computing the kinetic part, via the duality, as a ξ -fractal fraction of the static one is missing. This document takes up exactly that gap. It is the "further work" on the dynamic sector announced in Doc. 285.

2 The fixed foundation: the reversible Hilbert-space basis

Before developing the dynamic sector, let the fixed foundation it sits on be recorded. The FFGFT QM translation (Doc. 282) is closed in three pieces:

1. **The unambiguous Hilbert map** (Type III, Doc. 270): $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$, states = fiber-valued fields, QM operation = unitary parallel transport, entanglement = holonomy around cycles (Doc. 230), three masses $\leftrightarrow (M, r, \theta)$. A *bijection* – fully reversible.
2. **The time-free mass form**: mass sits on a separate tensor factor; CHSH = $2\sqrt{2}$ stays identical with and without the $\mathbb{C}^3$ fiber to $4 \cdot 10^{-16}$ – no static QM result changes. Time is *not computed* here; the reduced dynamics is time-local, unitary, Markovian.
3. **The time form**: adds exactly *one* effect – non-Markovian memory in the reduced dynamics (backflow $0 \rightarrow 0.53$, CP-divisibility violated) – which itself still lies within QM: a traced-out unitary bath reproduces it (Doc. 282/283). At the full level, therefore, unitary/reversible.

Reversibility is precise *per type* (Doc. 270): Type III (Hilbert) fully reversible; Type I (mass circle $\rightarrow$ time, "time taken into the space") *almost* reversible – lossless in mode content, only the winding number stays ambiguous; Type II (spatial projection $D3 \rightarrow D0$) *not* reversible. The only genuine irreversibility is the spatial projection – the measurement / decompactification (Doc. 270/284) –, not the time-inclusion. The non-Markovian memory is not a break of reversibility but the reduced view.

Aim of the dynamic sector. On this fixed, reversible foundation, the actual task is the *Hilbert-space extension and its translation: another description out of the Hilbert space* – the complementary, decompactified/projected side that maps computationally, via $D_f = 3 - \xi$ and ξ , onto SI units. The SI map is exactly the measurement/projection side (Type I/II), and D_f and ξ are the bridge quantities (Doc. 270/249/133). The dynamic sector is thus not a departure from the foundation but its unrolling into measurable units – and the sought kinetic ξ -fraction is one building block of exactly that unrolling.

.3 What is fixed

The duality fixes the *placement*, not the size. With $\tilde{T} = 1/\max(m, \omega)$ (Doc. 067, 080), an almost-massless window state has $\omega \gg m$, hence $\tilde{T} = 1/\omega$: the state sits in the *kinetic* sector. That is secure. So is the conceptual separation:

	static	kinetic
form	$E_0 = mc^2$ (rest)	$E = hf$ (free)
seat	Z_3 core (winding)	window, when a dynamics runs
time	$\tilde{T} = 1/m$	$\tilde{T} = 1/\omega$
status	derived (Doc. 285)	<i>open</i> (this document)

A distinction between two kinetic quantities matters: the ordinary kinetic energy of the electron, $(\gamma - 1)mc^2$ (Doc. 080), is *not* ξ -suppressed. The ξ -fractal fraction sought here is the *window rest-mass* of an otherwise massless state – not the kinetic energy of a massive one.

.4 The single data point: the neutrino

Doc. 047 treats the neutrino as a “near-massless photon”: $E^2 = (pc)^2 + (\sqrt{\xi^2/2} mc^2)^2$, with $v_\nu = c(1 - \xi^2/2) \approx 0.9999999911c$ and $m_\nu \sim \frac{\xi^2}{2} m_e$. This is the only existing value for a window rest-mass as a ξ -power of a core mass. It is, however, explicitly an *ansatz* (from the velocity/photon analogy), not *derived* from the duality. Doc. 204 additionally supplies a ξ -correction to the kinetic energy ($+0.69\xi_0$). There are no further data points; in particular, a single value is not enough for the Δm^2 spectrum.

.5 Narrowing the ξ -power: $p = 2$

With the ansatz $m_{\text{window}} = C \cdot \xi^p \cdot m_{\text{ref}}$, tested against the bounds (reproducing the $\xi^2/2$ value; the photon limit; dimensional consistency; an oscillation floor $\gtrsim \sqrt{\Delta m_{\text{atm}}^2} \approx 49$ meV; a direct kinematic bound $m_\nu \lesssim 0.8$ eV (KATRIN, model-independent)), only $p = 2$ is compatible with the meV scale: $p = 1$ gives $\mathcal{O}(10\text{--}70)$ eV (far above the bound), $p \geq 3$ falls below every anchor. So $p = 2$ is the *unique* order-of-magnitude compatible power – a genuine order-of-magnitude bound, not a fit.

Numerology warning. Within $p = 2$, the coefficient ($\frac{1}{2}$) and the reference mass are *not* fixed by the single anchor. Different references land somewhere in the neutrino window (e.g. $\xi^2/2 \cdot m_e \approx 4.5$ meV, $\xi^2 \cdot m_e \approx 9$ meV, $\xi^2/2 \cdot \sqrt{m_e m_\mu} \approx 65$ meV). This is *not* evidence but the P35 trap: coefficient and reference need a derivation, not tuning. Computed: `kinetik_xi_anteil.py`.

.6 The structural lead: $m_{\text{window}} = \xi \cdot \omega$

The Doc. 047 velocity expression $v_v = c(1 - \xi^2/2)$ is exactly equivalent to a ξ^1 relation. For $m \ll p$ one has $v/c = 1 - \frac{1}{2}(m/p)^2$; substituting gives $(m/p)^2 = \xi^2$, hence

$$m_{\text{window}} = \xi \cdot p \quad (\approx \xi \cdot \omega \text{ via } \tilde{T} \cdot m = 1). \quad (1)$$

The " ξ^2 " in the notation "mass = $\xi^2/2 \cdot m_e$ " is thus presumably the *square image* of a fundamental ξ^1 relation: the static rest-mass is ξ times the kinetic scale. This is the natural anchor for a derivation *via the duality*, because $\omega = 1/\tilde{T}$ is exactly the duality quantity. The open question therefore shifts from "why ξ^2 ?" to "why ξ^1 between rest-mass and kinetic scale, and what sets ω absolutely?".

.7 What remains to be derived

Three points are explicitly open:

1. **The reference/scale – largely closed** (next section). The two lab anchors select the reference as the *single* FFGFT scale $E_0 = \sqrt{m_e m_\mu}$; in its units $\omega/E_0 = \xi/2$ and $m_{\text{window}}/E_0 = \xi^2/2$. Residual: the *one* structural statement $\omega = \xi E_0$ (exactly one fractal ξ -power).
2. **The coefficient.** The factor $\frac{1}{2}$ in $\xi^2/2$ is not derived; possibly it is purely the velocity square ($\frac{1}{2}(m/p)^2$), i.e. no free factor at all.
3. **The spectrum.** A single ξ -fractal value sets the *scale*, not the *spectrum* (Δm_{sol}^2 , Δm_{atm}^2 , mixing angles). That demands the specified T^7 dynamics (the second open computation, Doc. 285).

.8 Closing the scale question: the lab anchors select E_0

The first open point – what sets ω absolutely – closes further than a free parameter would. In FFGFT there is only *one* absolute scale; everything else is dimensionless ratios. The question is therefore not "which number" but "which of the existing scales". With $m_{\text{window}} = \frac{\xi^2}{2} \cdot \text{reference}$, the two *lab* anchors (oscillation floor ≥ 49 meV; KATRIN ≤ 0.8 eV) bracket the reference:

reference	m_{window}	floor ≥ 49 meV	KATRIN ≤ 0.8 eV
m_e	4.5 meV	<i>fails</i> (too light)	yes
$E_0 = \sqrt{m_e m_\mu}$	65 meV	yes	yes
$M \approx 314$ MeV	2.8 eV	yes	<i>fails</i>
m_μ	940 meV	yes	<i>fails</i> (too heavy)

m_e is too light, M and m_μ too heavy – *only* $E_0 = \sqrt{m_e m_\mu}$ fits, and that is exactly the one natural FFGFT scale (Doc. 282, $\lambda_e \lambda_\mu = E_0$). The reference is thus not a free quantity but the existing scale, selected by the lab.

In units of E_0 everything then becomes a pure ξ -power times $\frac{1}{2}$:

$$\frac{\omega}{E_0} = \frac{\xi}{2}, \quad \frac{m_{\text{window}}}{E_0} = \frac{\xi^2}{2}. \quad (2)$$

No free parameter remains. The $\frac{1}{2}$ is the velocity square ($\frac{1}{2}(m/p)^2$, Doc. 047) – kinematic, not a tuning factor.

What honestly remains. The lab window (49–800 meV) is a factor $\sim$ few wide: it selects the *scale* E_0 sharply (excluding the alternatives by orders of magnitude) but does not pin the exact coefficient by itself. What remains is exactly *one* structural statement,

$$\omega = \xi E_0, \quad (3)$$

– E_0 suppressed by exactly *one* fractal ξ -power ($D_f = 3 - \xi$). This is no longer “free scale, self-referential” but a non-circular computation step: ω as the ξ -suppressed eigenvalue of the connection-Laplacian in the time-extended Hilbert structure (Doc. 283 §8). It closes the point – or falsifies it. Computed: `schliesse_omega.py`.

.9 The ξ^1 mechanism: lifting the zero mode by the dimension deficit

The remaining residual – why $\omega = \xi E_0$ with exactly *one* ξ -power – has a concrete mechanism. The connection-Laplacian on T^4 has an exact zero mode (the holonomy-flat constant). The fractal correction $g = \eta(1 + \xi f)$ acts as a diagonal potential perturbation and lifts it; first-order perturbation theory gives

$$\delta\lambda_0 = \xi \langle 0 | f | 0 \rangle = \xi \cdot \bar{f}, \quad (4)$$

i.e. the power ξ^1 *precisely when* the correction has a non-vanishing torus average $\bar{f} \neq 0$; for $\bar{f} = 0$ the first order vanishes and the mode lifts only at ξ^2 .

$D_f = 3 - \xi$ is a dimension *deficit* – a systematic, non-volume-preserving change of the measure –, hence $\bar{f} \neq 0$. So the power is *one*. Numerically (`xi1_eigenwert.py`): the zero-mode lift scales with $p = 0.999$ for a deficit ($\bar{f} \neq 0$) versus $p = 1.98$ for a volume-preserving ripple ($\bar{f} = 0$) – so the power is not accidental but follows from the nature of $D_f = 3 - \xi$.

Status. The ξ^1 power is thus *derived* (zero-mode lift by the dimension deficit), not asserted. What remains is only the $O(1)$ coefficient – kinematically the $\frac{1}{2}$ (Doc. 047) – and confirming that the physical kinetic state is this lifted zero mode. With that, $\omega = \frac{\xi}{2} E_0$ is closed up to an $O(1)$ factor.

.10 Relation to the static sector

The static sector is closed and computed: the Z_3 mass operator ($r = \sqrt{2}$, $\theta = 2/9$) plus the mass circle yields Koide $2/3$, the lepton ratios and – as a pre-registered, matched-null-tested declaration – the BD12-XCORE interface (Doc. 285). The dynamic sector is the *complementary, open* side: the same parameter ξ , but as a *fraction* rather than an *eigenvalue*. A derivation of $m_{\text{window}} = \xi \cdot \omega$ would move the window-filling dynamics (Doc. 263, holographic scale projection) from description to computation, and the neutrino-like $3'$ candidate (Doc. 285) from conditional to derived.

.11 Reproducibility

`kinetik_xi_anteil.py` (numpy-only): narrowing of the ξ -power against the bounds, the duality framing ($\tilde{T} = 1/\max(m, \omega)$), and the finding $p = 2$ with the explicitly open reference/coefficient/spectrum gap. The script claims no derivation; it narrows. `schliesse_omega.py`: the lab selection of the reference E_0 (table) and the reduction $\omega/E_0 = \xi/2$, $m_{\text{window}}/E_0 = \xi^2/2$, with the ξ^1 remainder as the marked closing point.